

ELECTROTECNIA Y MÁQUINAS ELÉCTRICAS



MÁQUINAS DE CORRIENTE CONTINUA

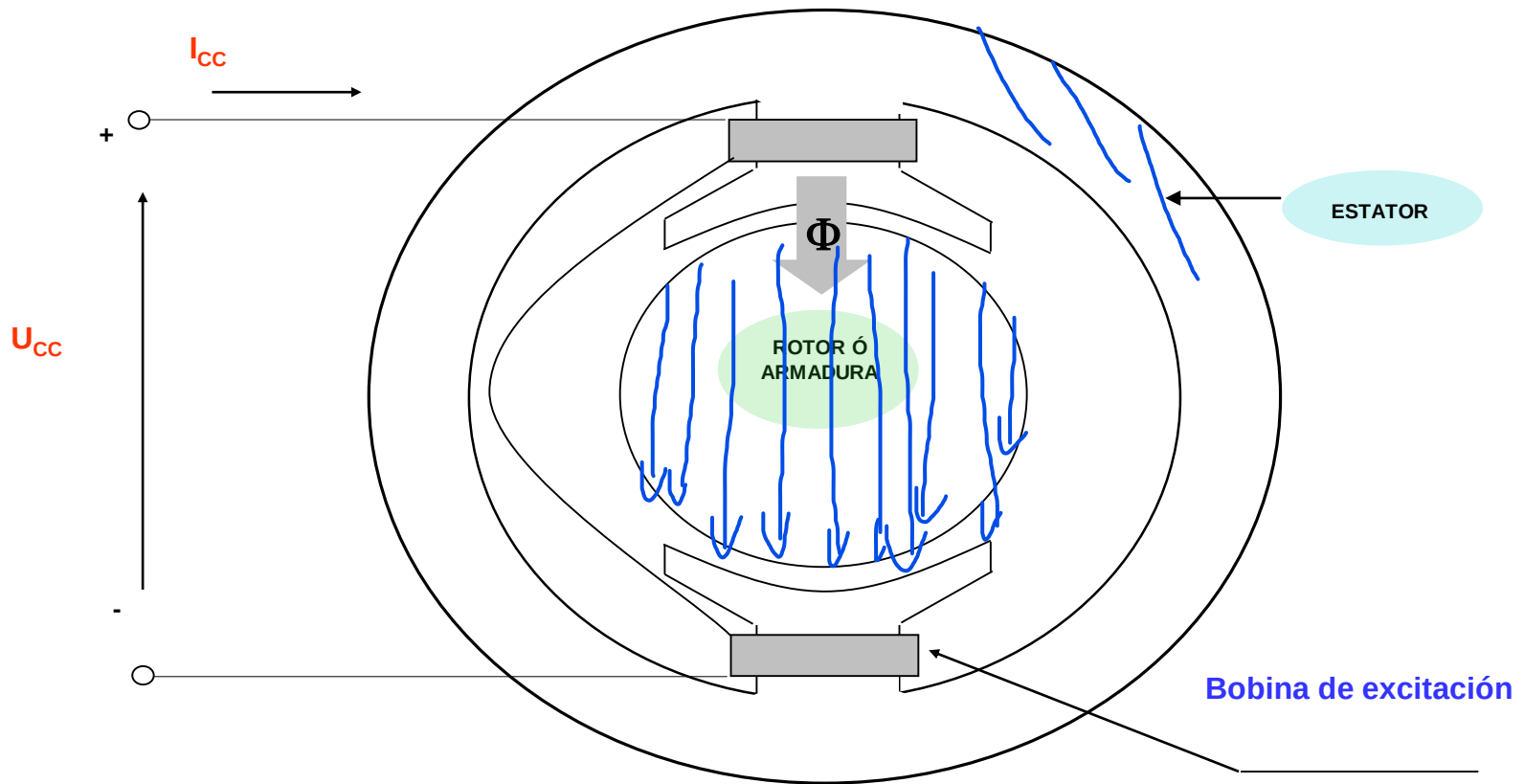
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CONCEPTOS GENERALES

Uso como generador y como motor. Principalmente motor.



CONCEPTOS GENERALES

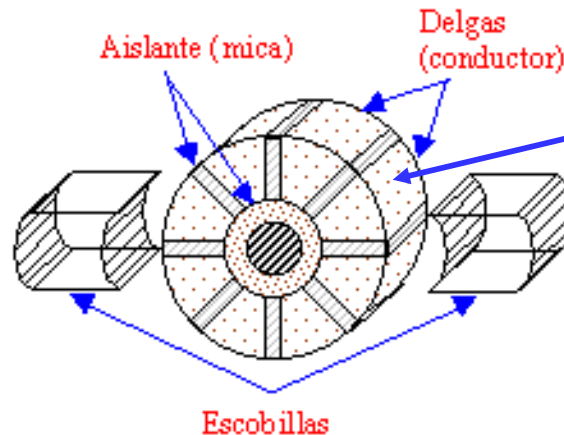
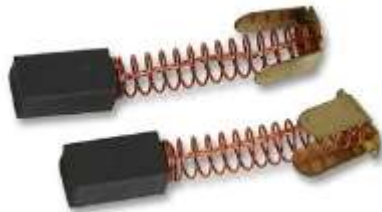
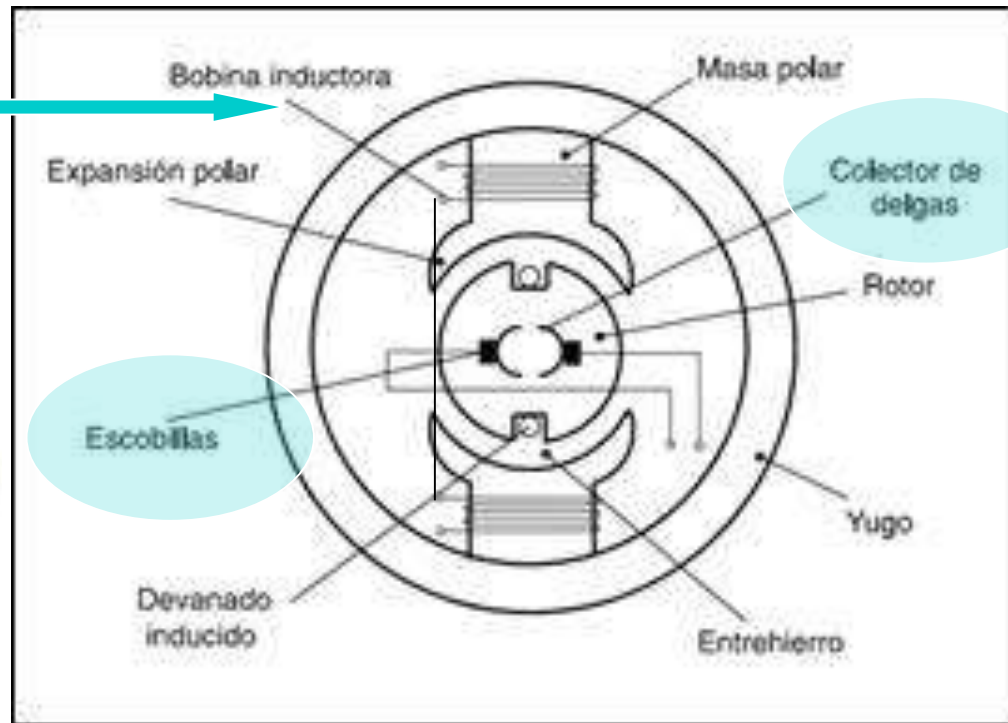


Estator o Carcaza, macizo, alimentado con cc.]

Rotor o Armadura, con chapas magnéticas, circula ca.

CONCEPTOS GENERALES

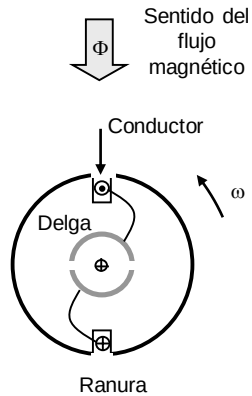
Icc o excitación



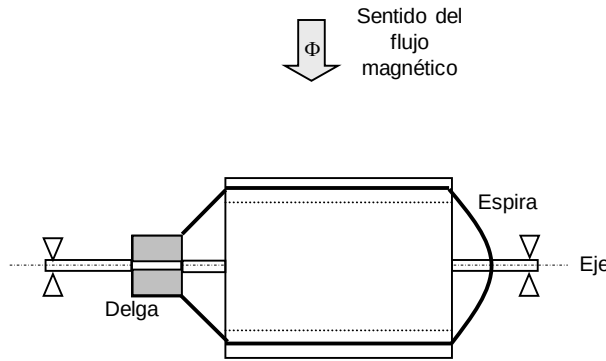
Colector o conmutador

GENERADOR

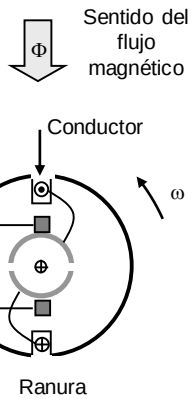
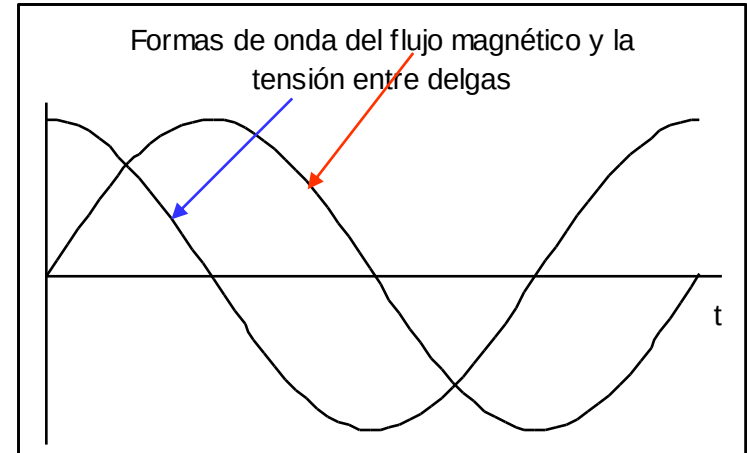
ANÁLISIS DOS DELGAS UNA BOBINA



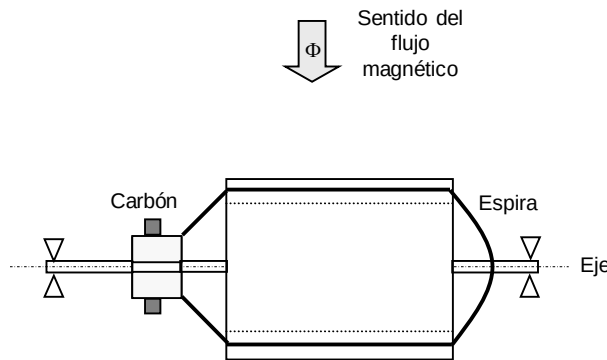
Vista frontal



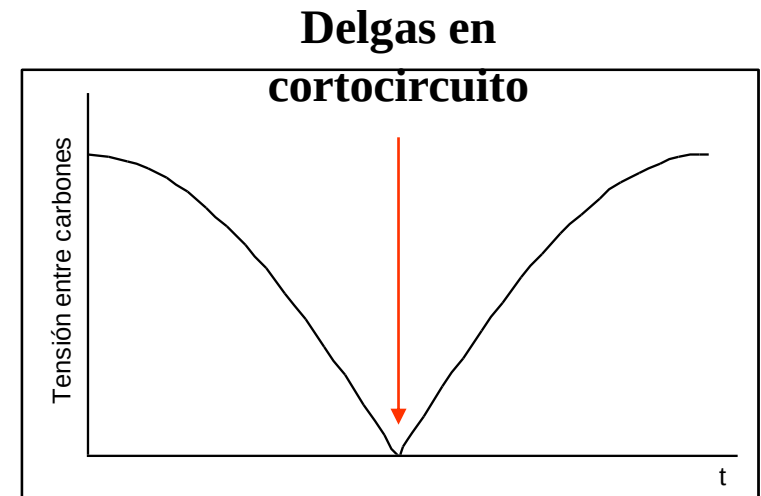
Vista lateral



Vista frontal

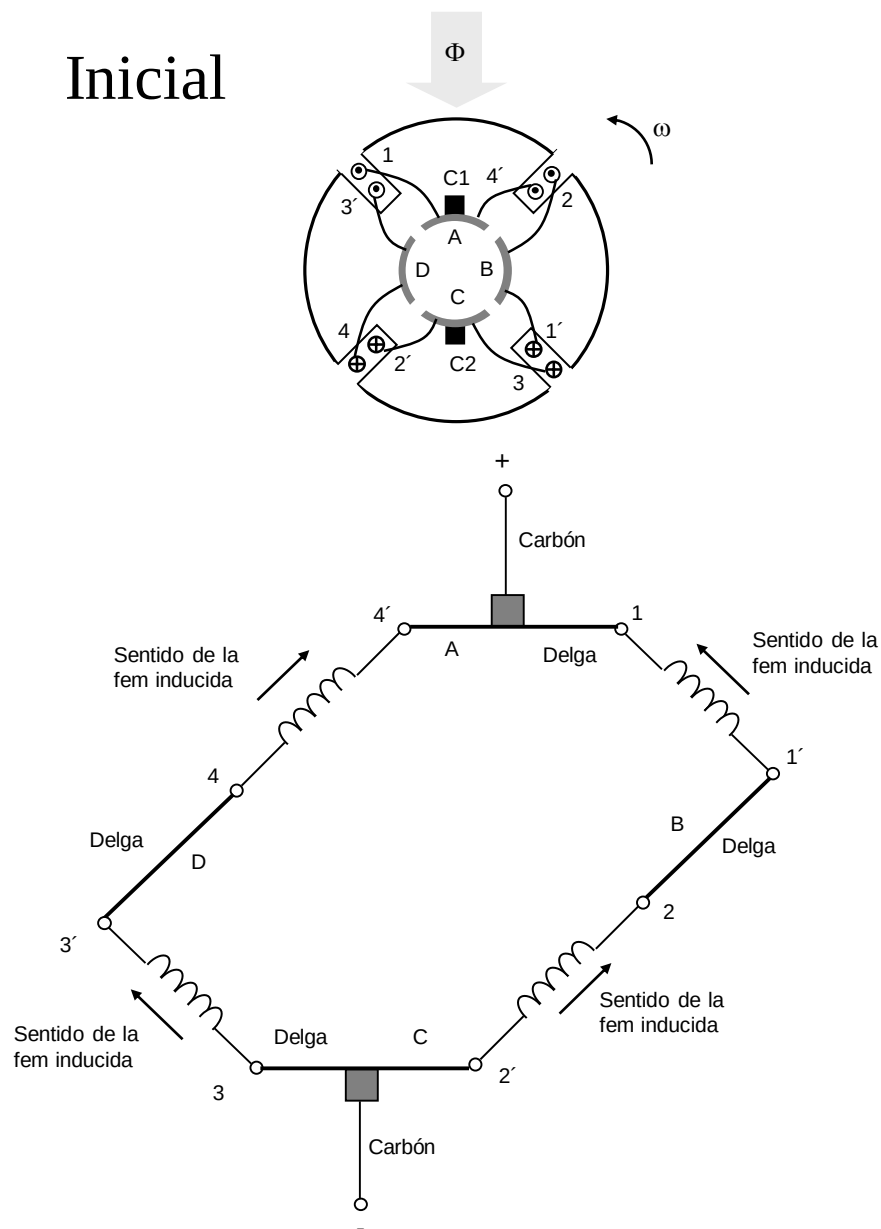


Vista lateral

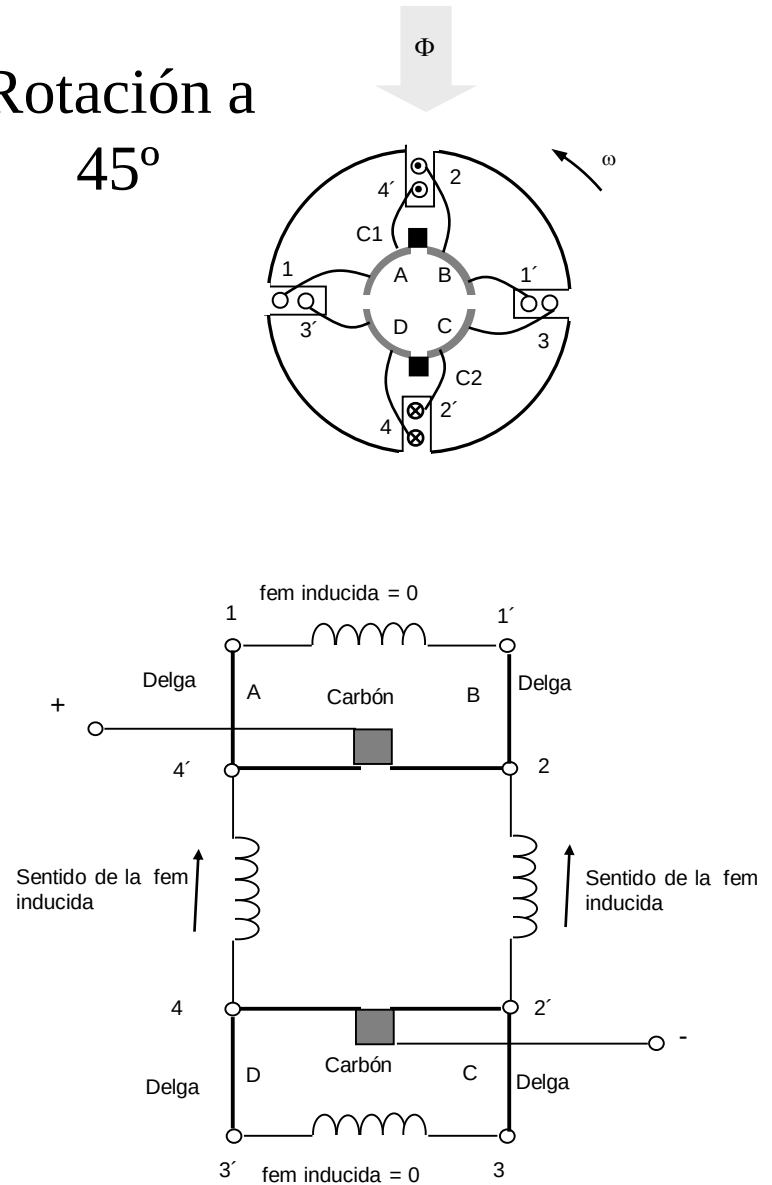


GENERADOR CON 4 DELGAS 4 BOBINAS

Inicial

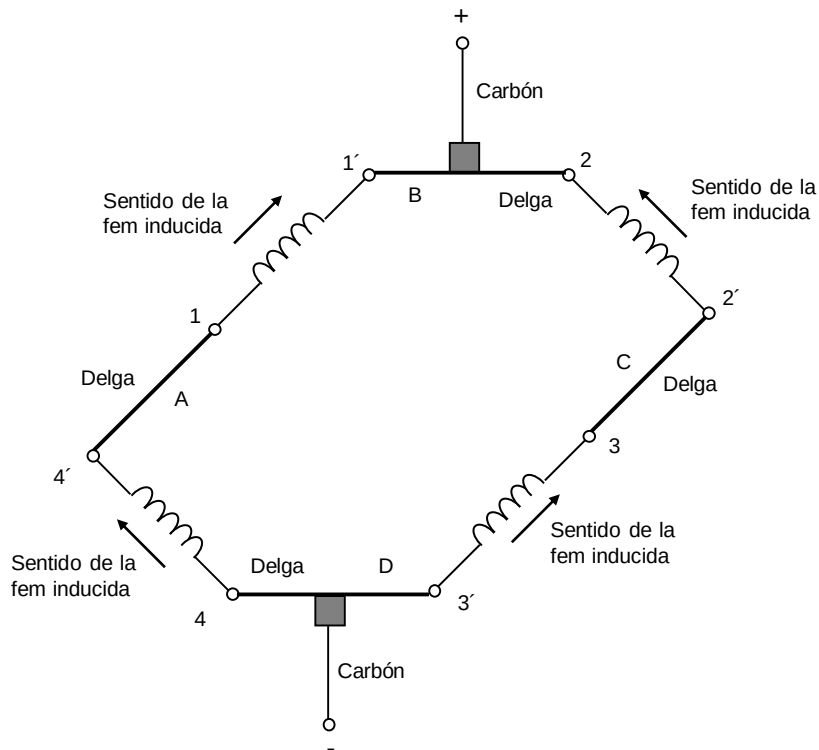
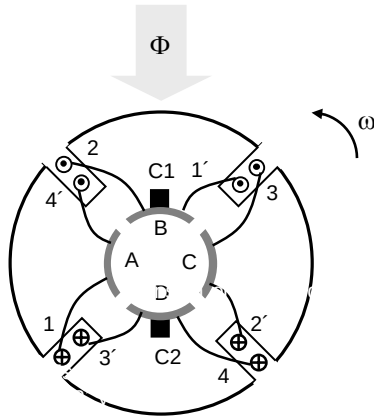


Rotación a
45°

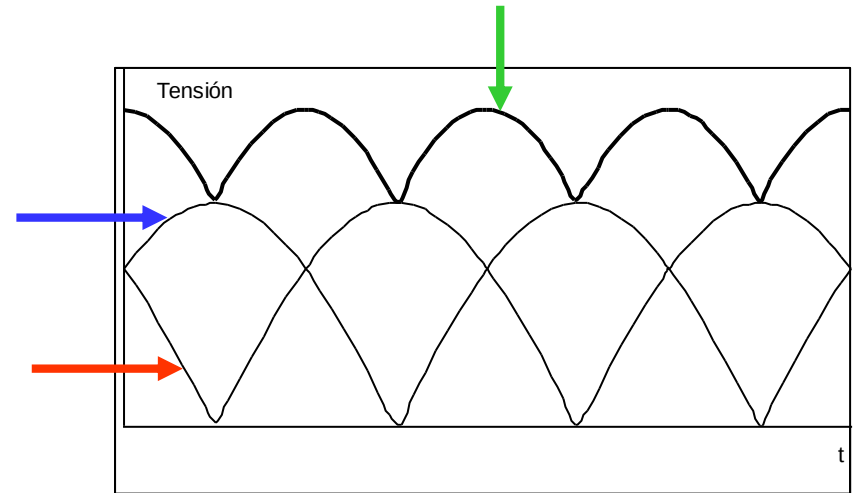


GENERADOR CON 4 DELGAS 4 BOBINAS

Rotación a
 90°



Onda resultante en
los carbonos

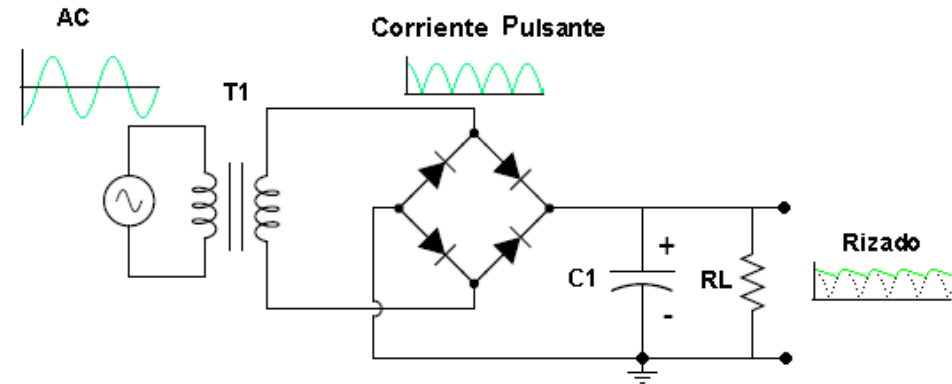
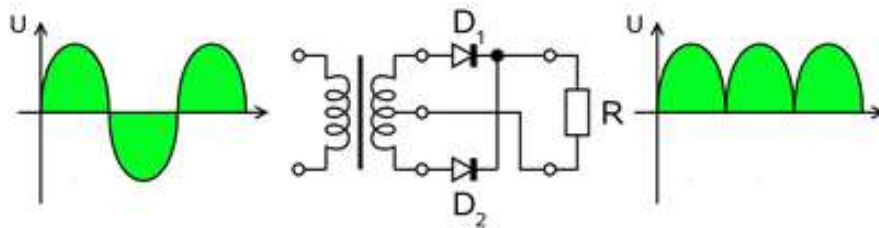
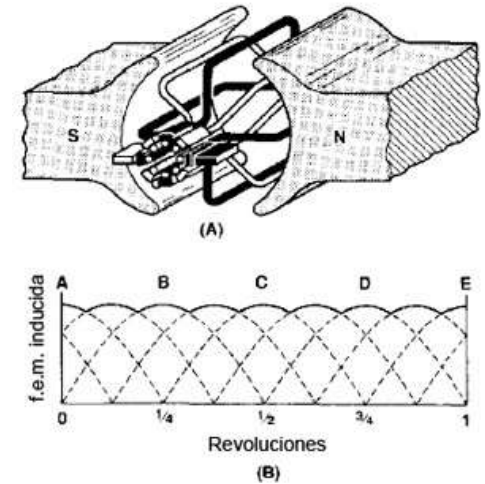
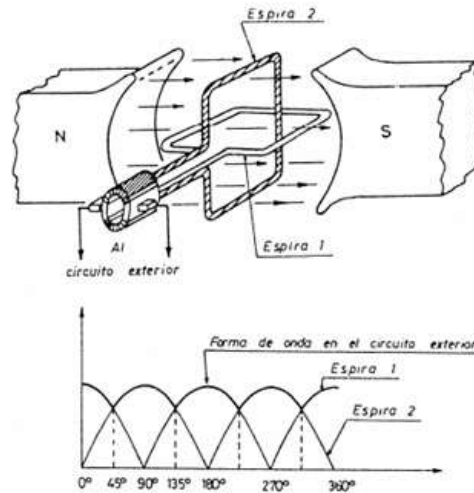
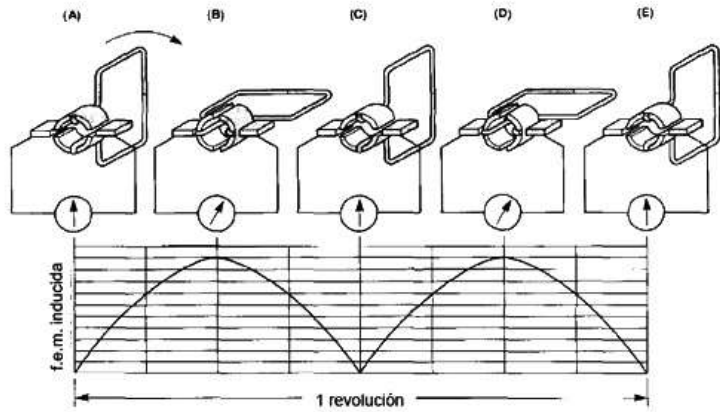


Onda rectificada en
las espiras

1-1' 3-3'

2-2' 4-4'

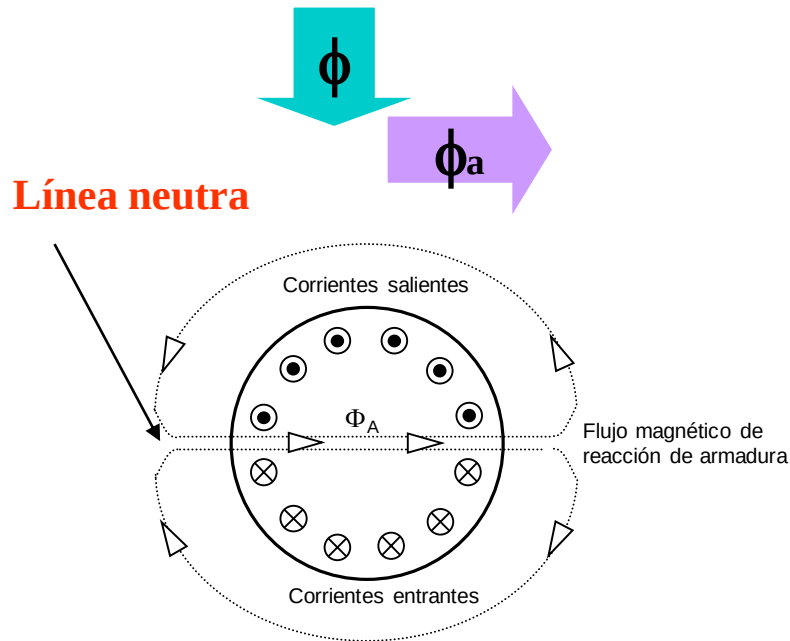
GENERADOR COMPARADO CON RECTIFICADOR



REACCIÓN DE ARMADURA O INDUCIDO

Flujo magnético

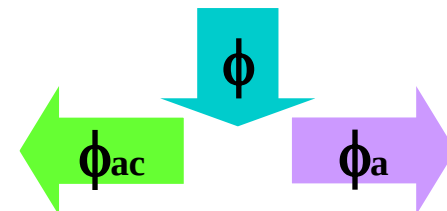
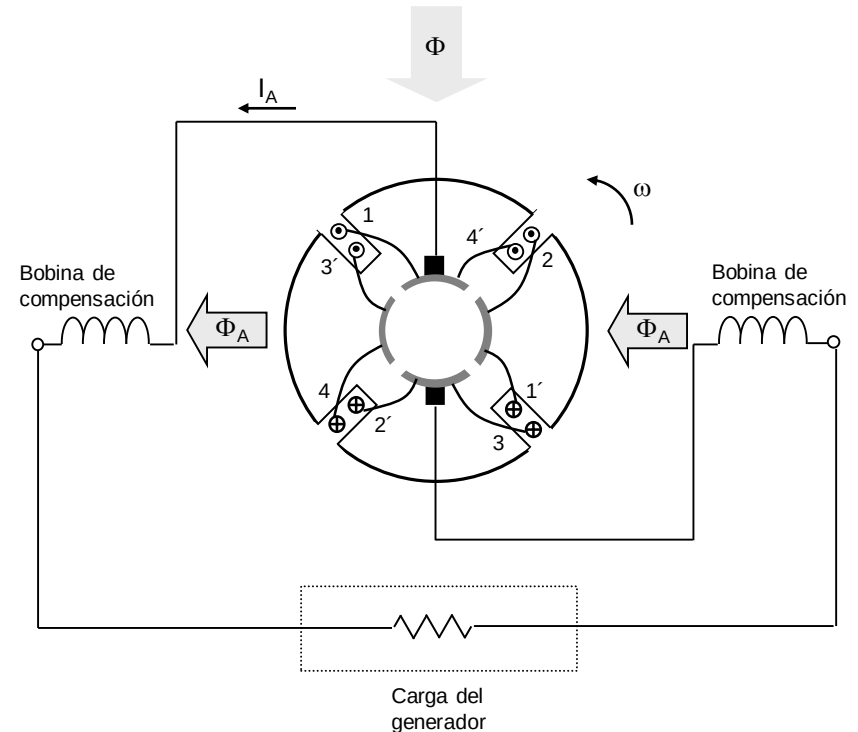
Reacción de armadura o rotor



Se produce al circular corriente cuando se pone una carga en el generador

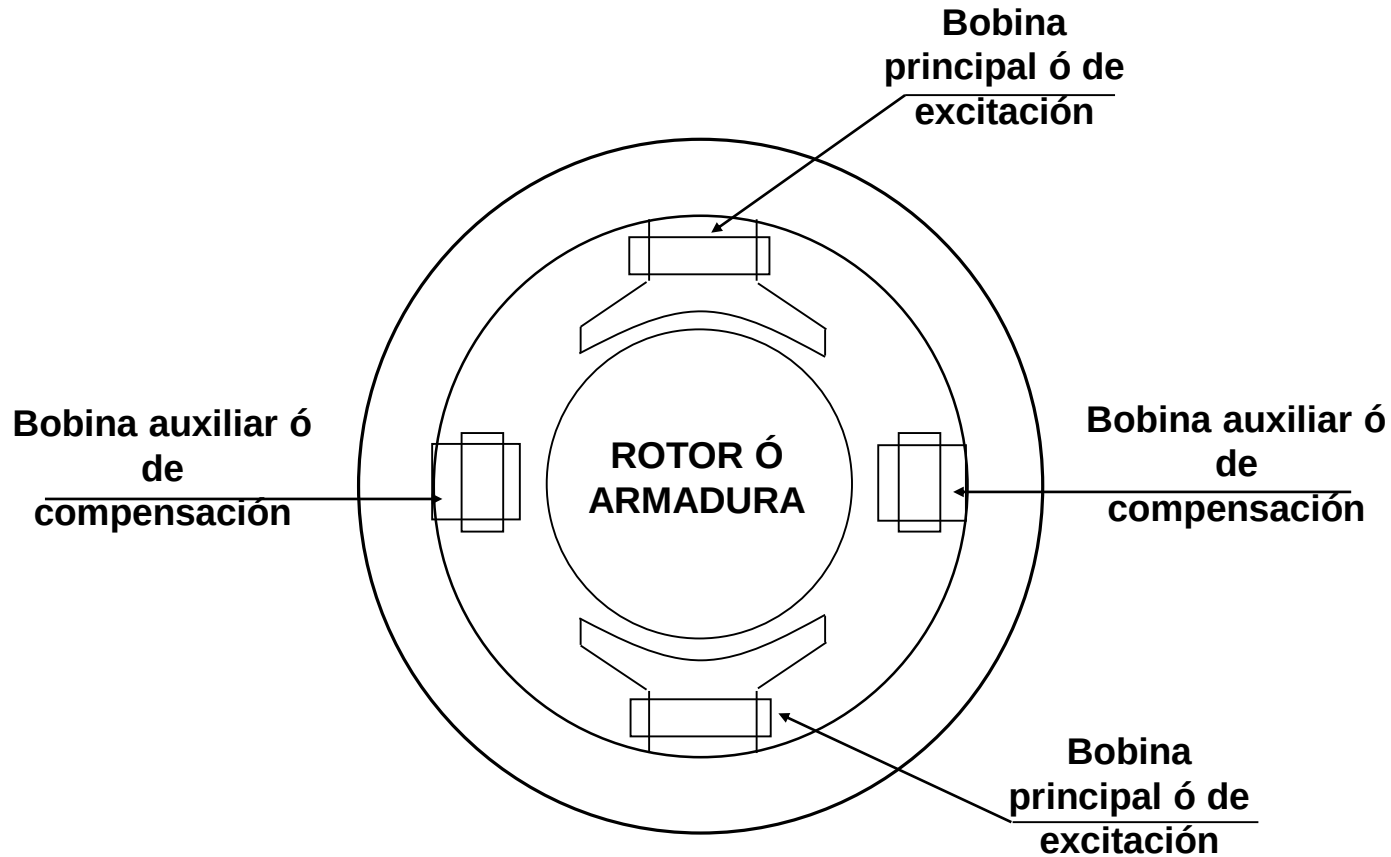
Flujo magnético

Compensación



REACCIÓN DE ARMADURA O INDUCIDO

Bobinas en el estator de la máquina cc



TENSIÓN INDUCIDA EN LOS CONDUCTORES DEL ROTOR (E)

$$E = B \cdot L \cdot v \quad [\text{V}]$$

$$B = \frac{\Phi}{S} = \frac{\Phi}{\pi D \cdot L} = \frac{p \cdot \Phi}{\pi \cdot D \cdot L}$$

p

**Inducción magnética por excitación
del estator**

$$v = 2 \cdot \pi \frac{n}{60} \frac{D}{2}$$

**Velocidad tangencial
del rotor**

$$E = \frac{P}{60} \cdot \Phi \cdot n$$

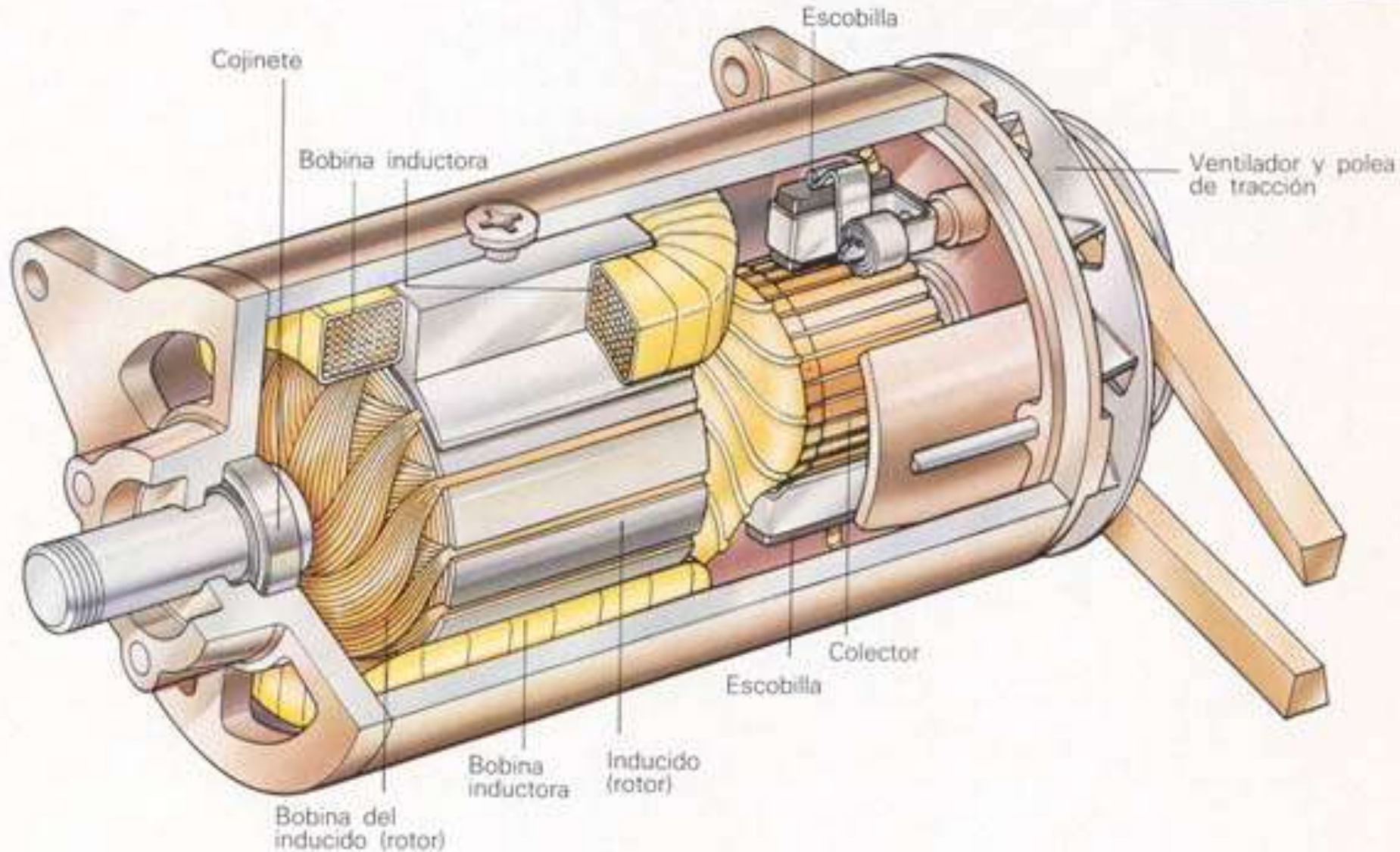
$$E = \frac{Z \cdot p}{a \cdot 60} \cdot \Phi \cdot n$$

K_1

“Z” conductores

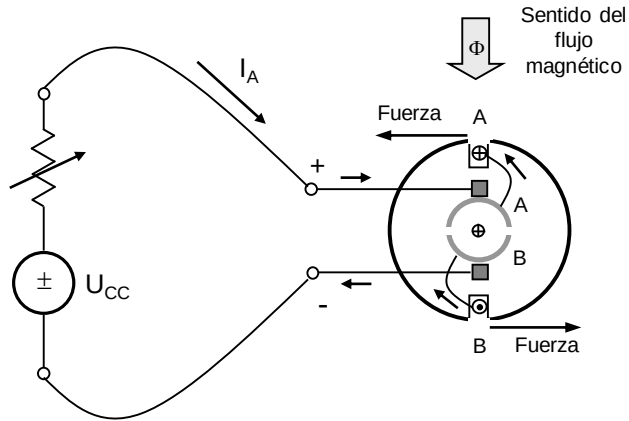
“a” ramas paralelo
divide I

GENERADOR DE CORRIENTE CONTINUA



1. Vista en sección de una dinamo típica. El esquema sencillo del inducido elemental (véase texto) ha sido reemplazado por unas bobinas para generar mucha energía. Por consiguiente, el conmutador tiene muchos segmentos de contacto.

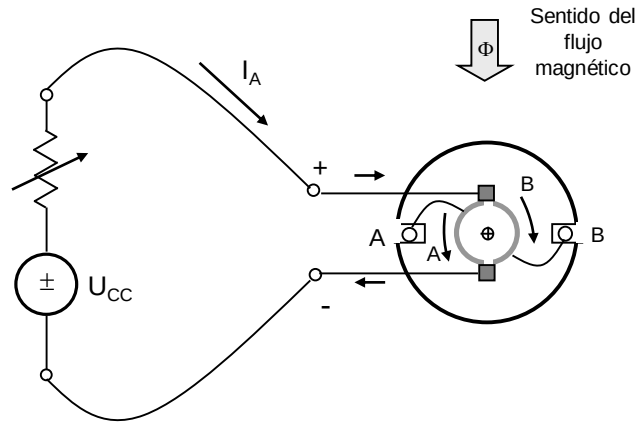
MOTOR DE CC



Circula I por espira.

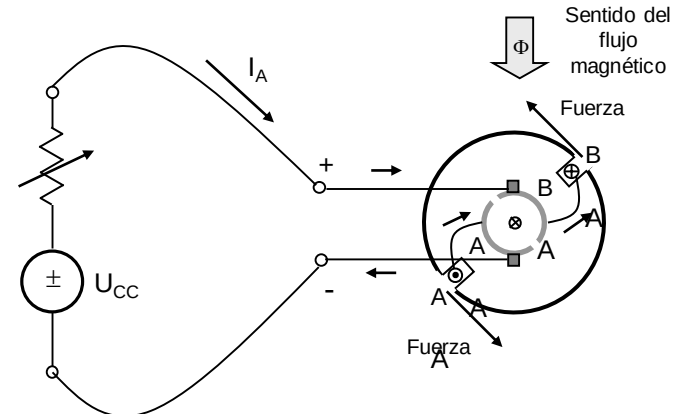
$$F = B \times L \times I$$

Cupla. Rotación.



No circula I por espira.

No cupla.



Situación
UNA espira

Si N espiras se mantiene el movimiento

CUPLA INTERNA DE LOS CONDUCTORES DEL ROTOR

$$T_{ic} = B \cdot I_c \cdot L \cdot D/2 \quad [\text{N.m}]$$

$$B = \frac{\Phi}{S} = \frac{\Phi}{\pi D \cdot L} = \frac{p \cdot \Phi}{\pi \cdot D \cdot L}$$

p

$$I_c = I_a / a$$

Inducción magnética por excitación del estator

Corriente por conductor del rotor

$$T_{ic} = \frac{P}{2 \cdot \pi \cdot a} \cdot \Phi \cdot I_a$$

$$T_i = \frac{Z \cdot p}{2 \cdot \pi \cdot a} \cdot \Phi \cdot I_a$$

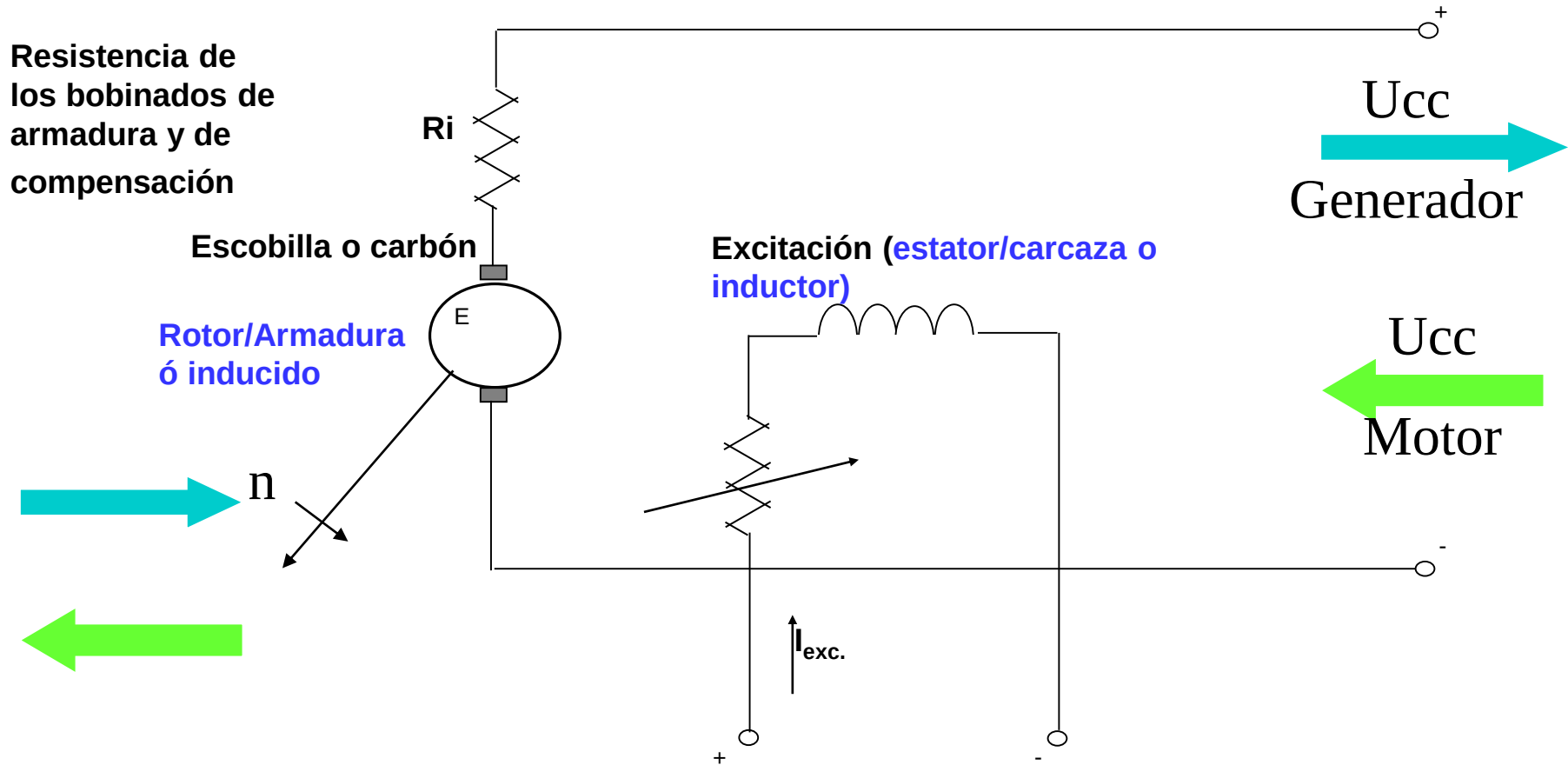
K

“Z” conductores

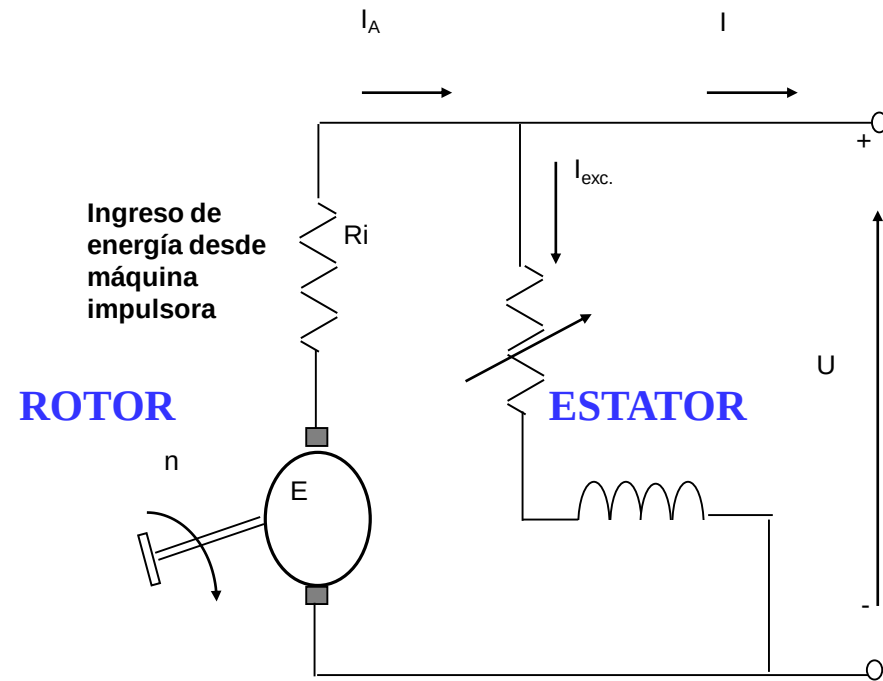
“a” ramas paralelo divide I que alimenta rotor



CIRCUITO EQUIVALENTE DE MÁQUINA DE CC

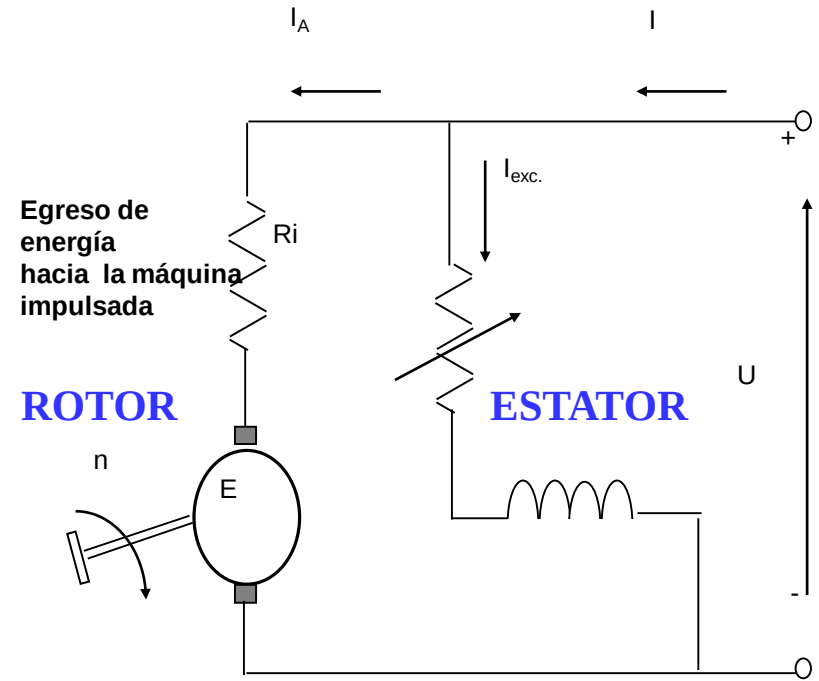


AUTOEXCITACIÓN EN DERIVACIÓN



Generador

$$U = E - I_a R_i$$

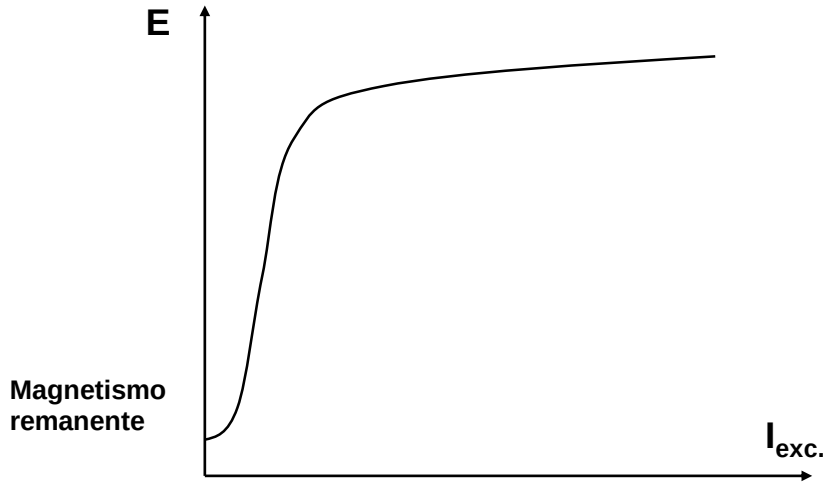


Motor

$$E = U - I_a R_i$$

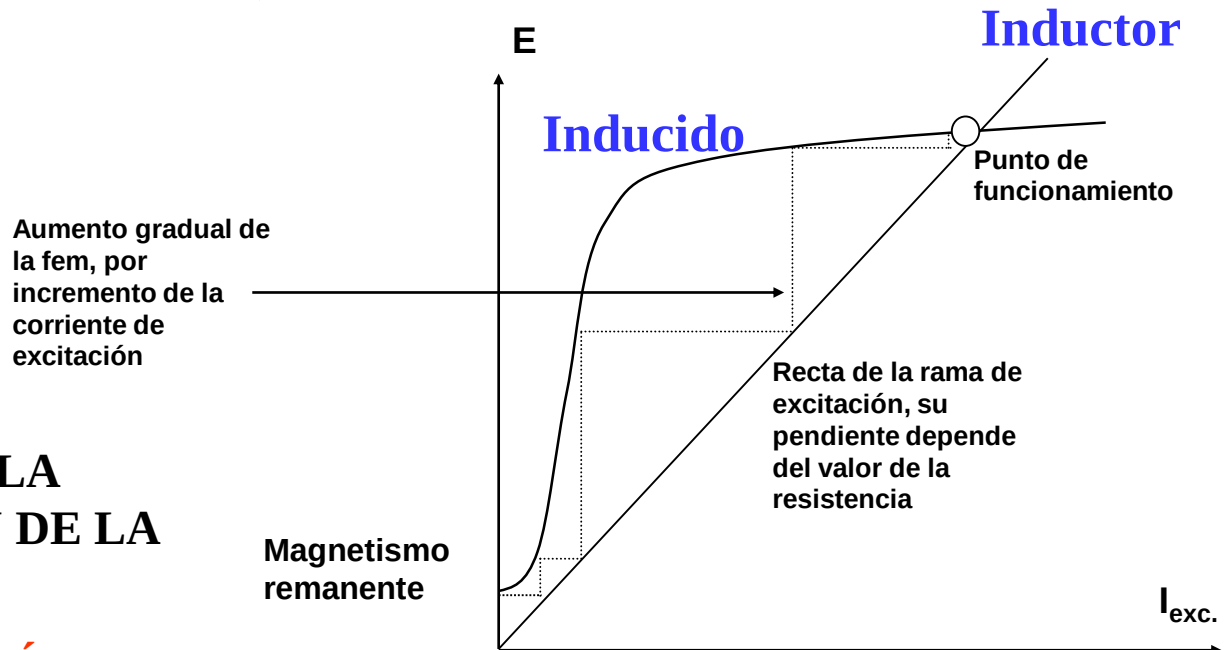
Agregar la caída de tensión por cada escobilla (se estima en 1 V)

FUNCIONAMIENTO DE UN GENERADOR DE CC AUTOEXCITADO EN DERIVACIÓN

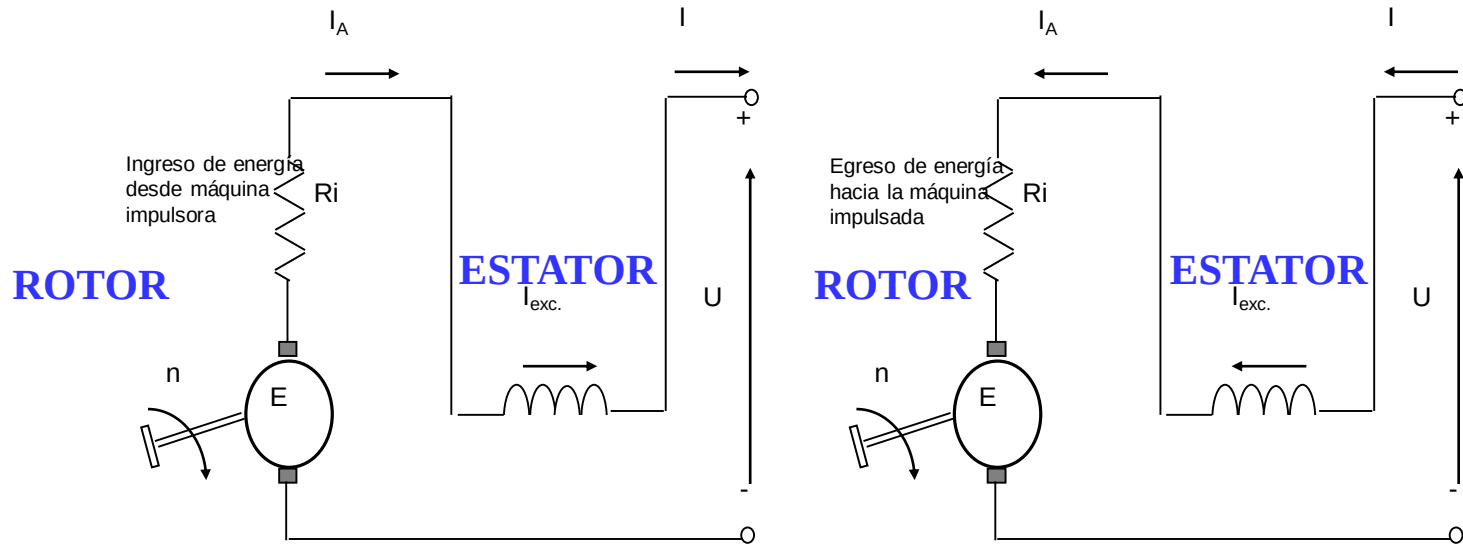


**IMPORTANCIA DE LA
CORRECTA CONEXIÓN DE LA
EXCITACIÓN**

DESMAGNETIZACIÓN



AUTOEXCITACIÓN EN SERIE



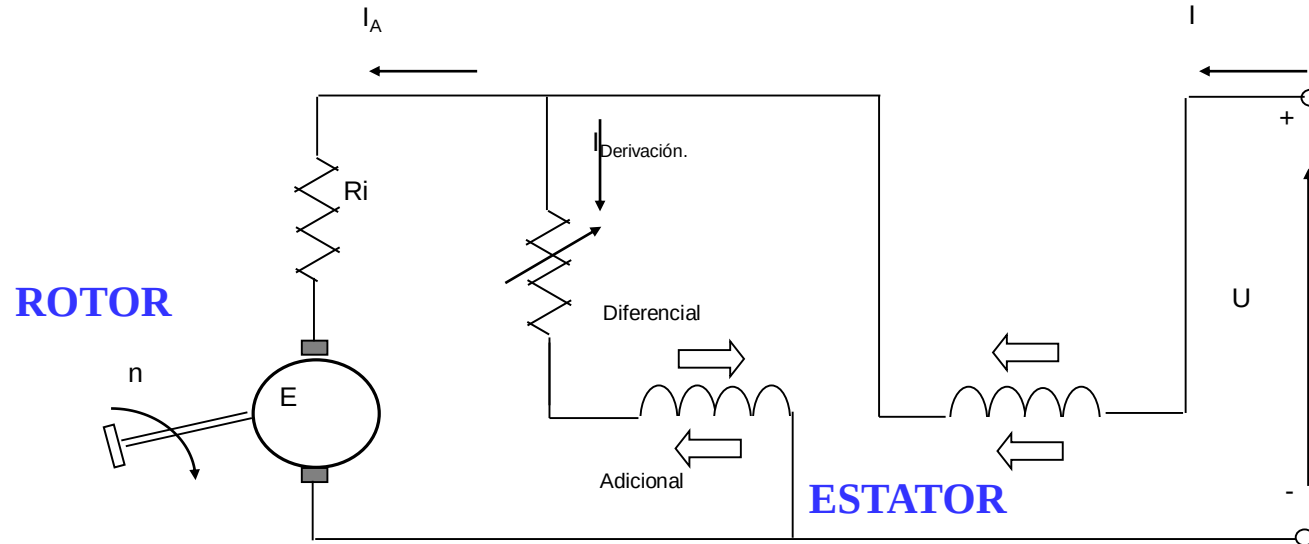
Generador

$$U = E - I_a (R_i + R_{exc})$$

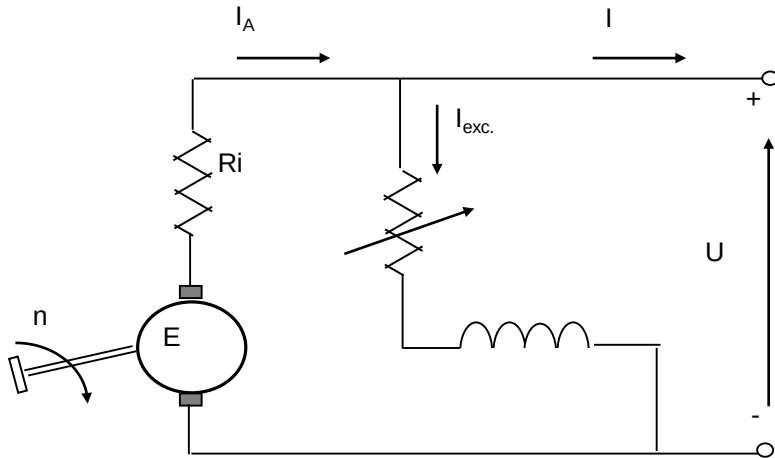
Motor

$$E = U - I_a (R_i + R_{exc})$$

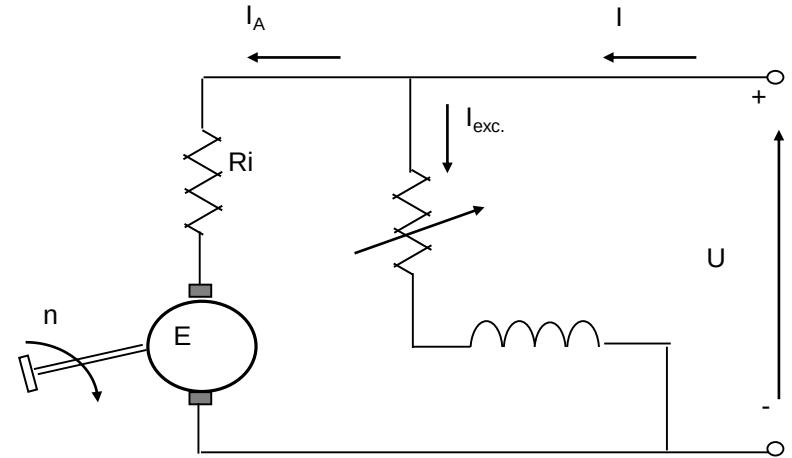
AUTOEXCITACIÓN COMPUESTA MOTOR



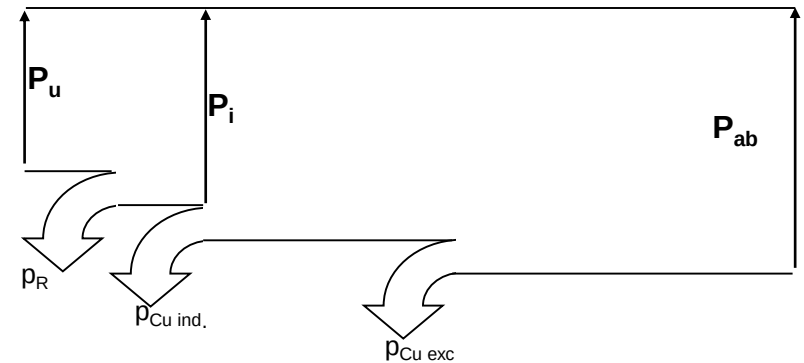
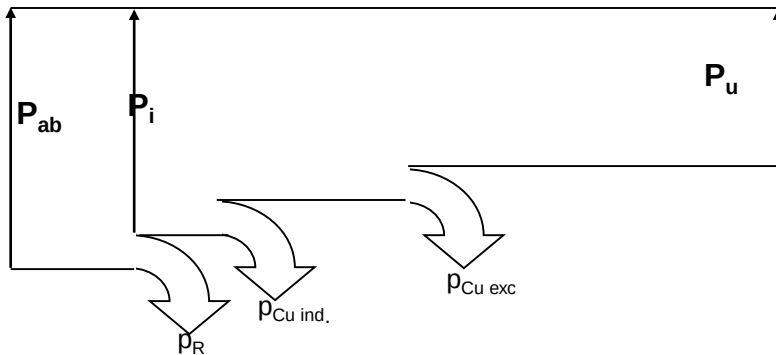
FLUJO DE POTENCIA



Generador



Motor



P_r (pérdidas rotacionales) = frotamiento y en el hierro

$P_{cu \text{ excitatriz}}$

$P_{cu \text{ inducido}}$

CUPLA Y VELOCIDAD EN MOTORES

Excitación en derivación

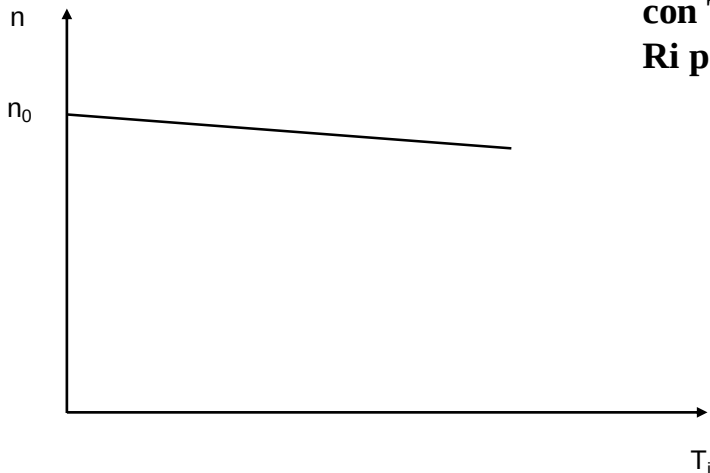
$$U = E + I_A \cdot R_i$$

$$E = K_1 \cdot \phi \cdot n$$

$$U = K_1 \cdot \phi \cdot n + I_A \cdot R_i \quad T_i = K \cdot \phi \cdot I_A$$

$$n = \frac{U}{K_1 \cdot \Phi} - \frac{R_i}{K \cdot K_1 \cdot \Phi^2} T_i$$

$$n = n_0 - \frac{R_i}{K \cdot K_1 \cdot \Phi^2} T_i$$



Poca variación
con T_i por ser
 R_i pequeño

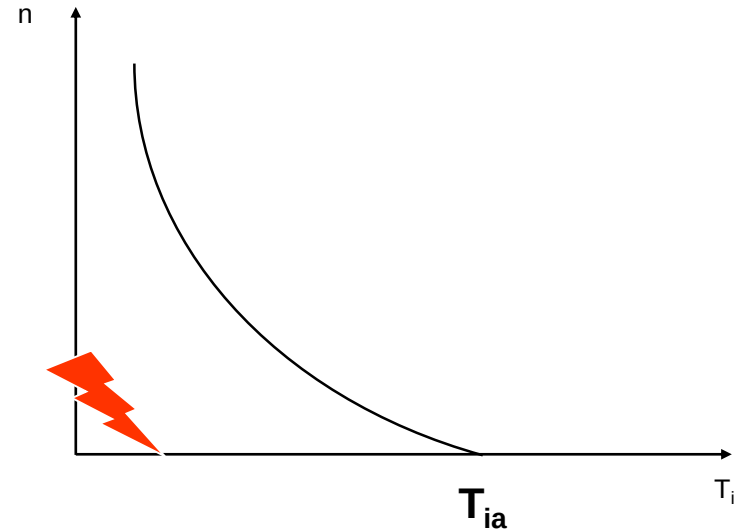
Excitación en serie

$$U = E + I_A \cdot (R_i + R_{exc})$$

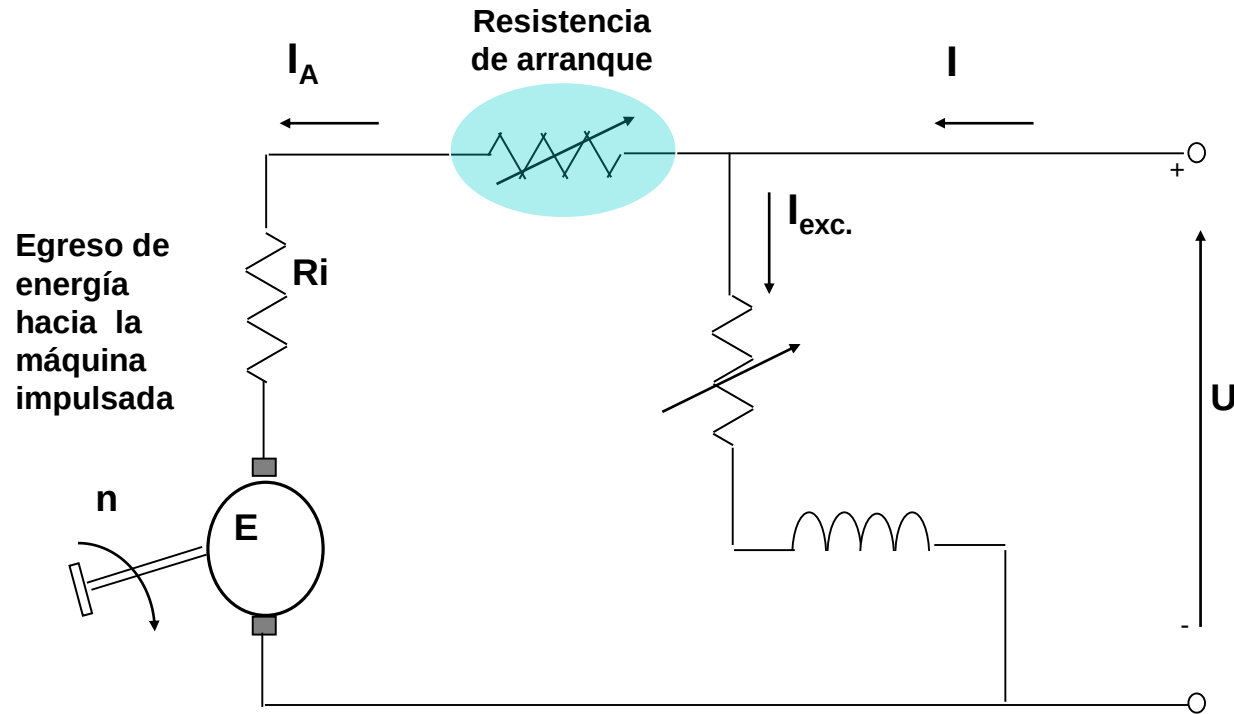
$$\phi = K_2 \cdot I_A$$

$$T_i = K \cdot \phi \cdot I_A$$

$$n = \frac{U}{\sqrt{\frac{T_i}{K \cdot K_2}} \cdot K_1 \cdot K_2} - \frac{(R_i + R_{exc})}{K_1 \cdot K_2}$$



CORRIENTE DE ARRANQUE EN MOTORES



$$U = E + R_i \cdot I_A$$

$$I_A = \frac{U - E}{R_i}$$

$$I_{A \text{ arranque}} = \frac{U}{R_i}$$

$$I_{A \text{ arranque}} = \frac{U}{R_i + R_{\text{arranque}}}$$